

A Sampling-Rate-Independent Physics-Informed Neural Network for Series Arc-Fault Detection in Switching Power Supplies Using Current Measurements Only

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Abstract—Series arc faults in switching power supplies (SMPS) are a leading ignition source in low-voltage installations. However, existing detectors either require voltage measurements, assume a known sampling rate, or embed predetermined arc equations whose validity on the target hardware is not demonstrated. To address this, a current-only, voltage-free detection framework is proposed in which the dynamic model is discovered from data rather than assumed. A cycle-scale device model of the nominal SMPS input stage is identified using the sparse identification of nonlinear dynamics (SINDy) algorithm, yielding a stable linear contraction map on half-cycle states (conducted charge, peak current, and conduction width) that is consistent with the capacitor charge-balance feedback of the diode-bridge rectifier input stage. The discovered model is verified independently through one-step prediction, multi-step rollout, null-dynamics controls, coefficient identifiability, and transfer to unseen SMPS models. The model is then embedded into a physics-informed neural network (PINN) through residual channels and a physics forecast loss. A causal period estimator removes all dependence on the sampling rate. The framework was evaluated under a nested leave-one-SMPS-model-out protocol with a one-shot outer test on a 24-recording arc-fault dataset comprising four SMPS models. The numerical results demonstrated the effectiveness of the proposed approach, achieving an outer-test average precision of 0.9924 without sampling-rate knowledge and 0.9954 with it, detecting 105/105 labelled arc events with a median delay of zero half-cycles and 0.59 false alarms per 100 cycles. Under resampling to 4.8 kHz and 3.2 kHz, the rate-independent detector retained an average precision of 0.984–0.985 with no rate information, whereas a -20% error in the assumed rate degraded the rate-aware detector to a Matthews correlation coefficient of 0.204. Matched ablations showed that the discovered dynamics improve detection when the learned representation is weak ($+0.073$ average precision, 95% confidence interval excluding zero) and are performance-neutral at the accuracy ceiling.

Index Terms—Arc-fault detection, physics-informed neural networks, SINDy, sampling-rate independence, switching power supplies, system identification.

I. INTRODUCTION

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This work was funded by Prince Sattam bin Abdulaziz University through project number (PSAU/2025/01/34748).

SERIES arc faults are among the most hazardous failure modes in low-voltage electrical systems, since the fault current remains below overcurrent protection thresholds while the localised arc plasma reaches temperatures sufficient to ignite surrounding material [1], [2]. The proliferation of switching power supplies (SMPS) in residential, commercial, and industrial installations aggravates this challenge, as the nonlinear input current of healthy SMPS loads resembles distorted arc signatures and thereby elevates both missed detections and nuisance tripping [3], [4]. Standardised arc-fault detection devices are governed by UL 1699B and IEC 62606 [5], [6]; however, the required sensitivity–selectivity trade-off across heterogeneous electronic loads remains an open problem [7], [8].

The detection techniques reported in the literature can be broadly classified into physics-based, signal-based, and learning-based approaches. Physics-based approaches employ dynamic arc conductance equations, most notably the Mayr [9] and Cassie [10] models, whose parameters are fitted to the observed waveforms. However, these equations were derived for high-power circuit-breaker arcs, and their identifiability from low-voltage terminal measurements is rarely examined. Signal-based approaches extract time-domain and frequency-domain features such as wavelet energies [11], [12], low-frequency harmonics and spectral signatures [13], [14], compressed-sensing representations [15], or multifeature fusion with heterogeneous ensembles [33]. These methods are effective, yet their thresholds are typically tuned to a fixed acquisition configuration. Learning-based approaches employ support vector machines [16], hidden Markov models [17], generative adversarial augmentation [18], and deep architectures under varying operating conditions [19], [20]; systematic surveys of the AI-driven detection landscape are provided in [3], [34]. Recently, physics-informed neural networks (PINNs) [21], [22] have been introduced into power system applications [23] and arc detection [24], [25], where an arc equation residual regularises the learned representation.

Despite the extensive research reported in the literature, three gaps persist. First, most dynamic-model-based detectors require the arc or branch voltage, which is rarely instrumented in field devices; current-only operation is therefore of practical importance [3]. Second, the sampling rate is almost universally assumed known and fixed, whereas deployed sensing platforms

exhibit heterogeneous and occasionally misreported acquisition rates; the sensitivity of detectors to such metadata errors is seldom quantified. Third, the physics embedded in existing PINN detectors is predetermined rather than demonstrated: the arc equation is imported from the circuit-breaker literature, and matched ablations isolating the contribution of the dynamics are generally absent. It is worth mentioning that a detector whose physics cannot be validated on the target hardware inherits the credibility of neither.

To address these gaps, this paper develops and faithfully validates a current-only, voltage-free, rate-independent dynamic PINN for arc-fault detection in SMPS. Rather than assuming an arc equation, the nominal *device* dynamics are discovered from data using the sparse identification of nonlinear dynamics (SINDy) algorithm [26]–[28], verified independently as a dynamic model, and only then embedded into the detector. Arcs are detected as violations of the validated nominal dynamics. The contributions of this work are:

- Discovery and independent verification of a cycle-scale dynamic model of the nominal SMPS input stage from current measurements only. The SINDy-discovered model is a stable linear contraction map on half-cycle states that is consistent with the capacitor charge-balance feedback of the diode-bridge rectifier topology, and it transfers to unseen SMPS models. Unlike existing PINN detectors that embed predetermined arc equations, the proposed framework demonstrates that arc-plasma equations (Mayr and Cassie forms) are unidentifiable from current-only terminal measurements, and it therefore models the device rather than the arc.
- A causal, rate-free processing pipeline in which the fundamental period is estimated online from trailing autocorrelation using no future samples and no rate metadata. The equation coefficients discovered by the rate-independent and rate-aware branches agree within approximately 2%.
- A leakage-free nested leave-one-SMPS-model-out evaluation with matched ablations, equal tuning budgets, per-fold frozen thresholds and calibration, and a one-shot outer test. The dynamics contribution is quantified honestly: it is statistically supported when the learned representation is weak and performance-neutral at the accuracy ceiling.
- Quantified robustness to acquisition-rate changes: the rate-independent detector is non-inferior to the rate-aware detector at the native rate and retains its accuracy under resampling with no rate information, whereas rate-metadata errors severely degrade the rate-aware alternative.

A. Motivation Example

The practical importance of rate independence can be illustrated using the outer-test system of this study. When the acquisition rate assumed by a conventional rate-aware detector deviates from the true rate by -20% , its Matthews correlation coefficient (MCC) at the frozen deployment threshold collapses from 0.925 to 0.204, and on one SMPS model to

-0.008 , i.e., the detector becomes uninformative. Similarly, a spectral-feature detector that dominated the inner validation folds collapsed to an average precision of 0.48–0.57 on the held-out hard SMPS model in the outer test. One can notice that static detectors tuned to a fixed acquisition configuration and a fixed device population can fail silently when either assumption is violated. Therefore, relying solely on rate-dependent signatures for monitoring heterogeneous SMPS populations leads to fragile detection, and it is necessary to design the processing pipeline to be independent of the sampling rate while validating generalisation on completely held-out device models.

II. PROBLEM FORMULATION

The objective is to decide, for each half mains cycle k of a measured SMPS input current, whether the device is operating normally or feeding a series arc, using the current samples only. Neither the branch voltage nor the sampling rate is available to the proposed detector. The overall framework comprises causal segmentation, half-cycle state extraction with a causal baseline, discovery and verification of the nominal device dynamics, and integration into a PINN detector under a nested evaluation protocol.

A. Dataset

A public SMPS arc-fault dataset comprising 24 recordings from four SMPS models (A–D) and two supplementary normal-control units is employed. Each recording contains 25,600 samples acquired at 6.4 kHz (4 s, 200 mains cycles, 128 samples per cycle). The signals are normalised 12-bit ADC counts; therefore, no ampere-valued physical parameters are claimed anywhere in this work. Expert labels mark no-arc and arc intervals per recording; the 256-sample gaps between labelled intervals correspond to ignition transitions and are excluded from all losses and metrics. In total, the labelled data comprise 21 arc-containing recordings with 105 labelled arc events across the outer evaluations, one no-arc-only recording (A4), and the two supplementary units, which are reserved strictly for evaluation. A label audit verified the indexing conventions, the transition-gap semantics, and the post-trip unlabelled regions before any modelling was performed; Fig. 1 shows a sample recording with its labels.

B. Causal Rate-Independent Segmentation

Two processing branches are implemented throughout. The *rate-aware* branch uses the known 128 samples per cycle. The *rate-independent* branch estimates the fundamental period causally from the trailing normalised autocorrelation of the buffered current. Let $\tilde{x}_m = x_m - \bar{x}$ denote the mean-removed samples of the trailing buffer of length $B = 1536$ ending at sample n . The autocorrelation at lag l is

$$\rho_n(l) = \frac{\sum_{m=l}^{B-1} \tilde{x}_{n-B+m} \tilde{x}_{n-B+m-l}}{\sum_{m=0}^{B-1} \tilde{x}_{n-B+m}^2}, \quad (1)$$

where l is searched over the design envelope $l \in [16, 512]$ samples and the estimate is updated every 64 samples. The raw

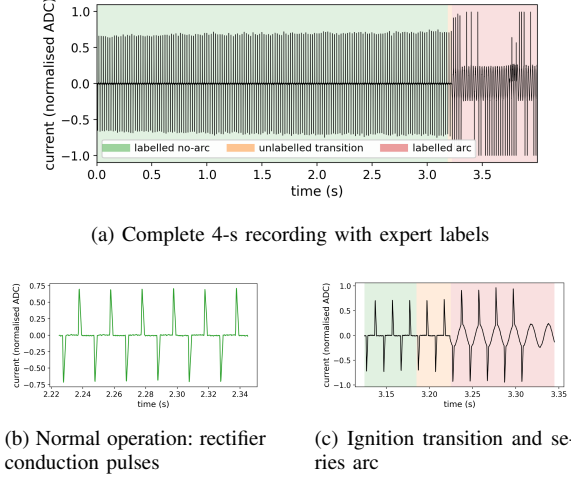


Fig. 1. Sample recording (Model C) at 6.4 kHz. Green: labelled no-arc; orange: unlabelled ignition transition (excluded from all losses and metrics); red: labelled arc. The nominal current consists of the periodic conduction pulses of the capacitor-input rectifier stage, whereas the arc interval exhibits amplitude collapse and irregular conduction.

maximiser $l^* = \arg \max_l \rho_n(l)$ can lock onto an integer multiple of the true period; therefore, a subharmonic disambiguation step replaces l^* by l^*/d for the largest divisor $d \in \{4, 3, 2\}$ satisfying

$$\rho_n(\lfloor l^*/d \rfloor) \geq 0.85 \rho_n(l^*), \quad (2)$$

i.e., a divisor lag is accepted only if it carries at least 85% of the dominant correlation. Sub-sample resolution then follows by parabolic interpolation around the selected lag,

$$\hat{P}(n) = l^* + \frac{1}{2} \frac{\rho_n(l^*-1) - \rho_n(l^*+1)}{\rho_n(l^*-1) - 2\rho_n(l^*) + \rho_n(l^*+1)}. \quad (3)$$

No future samples and no rate variable enter (1)–(3); the lag bounds constitute a design envelope analogous to a sensor bandwidth rather than rate knowledge. Half-cycle boundaries then advance causally by $\hat{P}/2$ samples from the end of the warm-up buffer. The comparison between the two branches, carried through every experiment, quantifies the cost of withholding the sampling rate.

C. Half-Cycle States and Causal Baseline

For each half cycle k , spanning N_k samples $\{i_1, \dots, i_{N_k}\}$ of the median-centred current, three states are computed from the rectified waveform:

$$\begin{aligned} Q_k &= \frac{64}{N_k} \sum_{m=1}^{N_k} |i_m|, & PK_k &= \max_m |i_m|, \\ W_k &= \frac{1}{N_k} \sum_{m=1}^{N_k} \mathbb{I}[|i_m| > 0.12 PK_k], \end{aligned} \quad (4)$$

where Q_k is the conducted charge normalised by the half-cycle length (the factor 64 references both branches to a common per-half-cycle scale), PK_k is the absolute peak, and W_k is the conduction-width fraction, i.e., the fraction of samples exceeding 12% of the half-cycle peak. The length normalisation in (4) is precisely what renders the states, and hence the dynamics discovered on them, independent of the

number of samples per cycle. Let $x_k = (Q_k, PK_k, W_k)^\top$. A training-data audit showed the operating point of every recording drifting slowly from its initial calibration (median growth $2.1\times$, 20/20 no-arc recordings), consistent with thermal warm-up; deviations are therefore defined against a *causal trailing median baseline*,

$$z_k = \frac{x_k - \text{med}\{x_{k-60}, \dots, x_{k-1}\}}{\sigma_{\text{cal}}}, \quad (5)$$

where $\text{med}\{\cdot\}$ is the elementwise trailing median over the previous 60 half cycles and σ_{cal} contains floored robust scales computed from the first 30 no-arc half cycles of the recording,

$$\begin{aligned} \sigma_{\text{cal}} &= \max(\text{std}_{\text{cal}}, 0.05 |\mu_{\text{cal}}|, s_{\text{floor}}), \\ s_{\text{floor}} &= (10^{-3}, 10^{-3}, 10^{-2})^\top, \end{aligned} \quad (6)$$

with the maximum taken elementwise, and std_{cal} and μ_{cal} the calibration standard deviation and mean. The floors in (6) are necessary because the conduction-width state is quantised to multiples of $1/N_k$ and its calibration standard deviation can approach zero, which would otherwise inflate the standardised deviations unboundedly. All quantities in (5)–(6) are causal and label-free at inference time.

D. Discovery of the Nominal Device Dynamics

The nominal dynamics of z_k are discovered using SINDy with sequentially thresholded least squares (STLSQ) [26]. It is worth mentioning that the discovered equation represents the nominal SMPS dynamics only; arc dynamics are deliberately not modelled, and arcs are expected to violate the nominal equation. The essential steps are given as:

Step 1: Assemble the regression pairs $(z_k, \Delta z_k)$, $\Delta z_k = z_{k+1} - z_k$, using only no-arc pairs from the training SMPS models. Arc labels never enter equation discovery or selection.

Step 2: Construct the candidate library

$$\Theta(z) = [1, Q, PK, W, Q^2, PK^2, W^2, QPK, QW, PKW], \quad (7)$$

containing constant, linear, and quadratic candidate terms of the state.

Step 3: Solve the sparse regression

$$\Delta z_k = \Theta(z_k) \Xi, \quad \Xi \in \mathbb{R}^{10 \times 3}, \quad (8)$$

for the coefficient matrix Ξ by STLSQ: an ordinary least-squares fit is followed by iteratively zeroing all coefficients with $|\xi_{ij}| < 0.05$ and refitting the surviving active set, repeated until the support stabilises. Each column of Ξ governs one state increment $(\Delta Q, \Delta PK, \Delta W)$.

Step 4: Select the library complexity by a label-free criterion (one-step no-arc prediction error on inner-validation models). The linear library was selected in all eight outer-fold-by-branch selections; the quadratic terms were rejected by the data.

The discovered equation, with bootstrap-mean coefficients agreeing across both branches within approximately 2%, is the linear contraction map

$$z_{k+1} = z_k + c + A z_k, \quad A \approx \begin{bmatrix} -0.46 & +0.01 & 0 \\ +0.06 & -0.83 & +0.01 \\ -0.17 & +0.10 & -0.34 \end{bmatrix}, \quad (9)$$

with $c \approx (0, +0.13, -0.07)^\top$. All eigenvalues of $I + A$ lie strictly inside the unit circle. Physically, (9) is the linearised charge-balance feedback of the capacitor-input rectifier stage: excess conducted charge raises the DC-link voltage, which suppresses the charge, peak, and conduction width of the subsequent half cycle. The rectifier-plus-capacitor topology itself was confirmed at the waveform level (normalised RMSE 0.04–0.18 on all no-arc recordings against 0.61–0.83 for a resistive control). Equation (9) is a SINDy-discovered, rectifier-informed adaptation; it is not a reproduction of a published equation.

E. Rejection of the Arc-Physics Route

It is important to report why the conventional arc-physics route was rejected, since these negative results are retained by design and motivate modelling the device instead of the arc. First, the Mayr conductance equation $\tau d(\ln g)/dt = iv/P_0 - 1$ requires the arc voltage v . Eliminating $v = i/g$ via Ohm's law and normalising the conductance and current by their characteristic scales, $z = g/G$ and $j = i/\sqrt{P_0 G}$, with dimensionless time $\theta = t/T_{\text{cyc}}$, yields the current-only form

$$\hat{\tau} \frac{dz}{d\theta} = j^2 - z, \quad (10)$$

where $\hat{\tau} = \tau/T_{\text{cyc}}$ is the normalised arc time constant. The reduction itself is exact: synthetic verification recovered the rescaled dynamics to 2.2×10^{-6} and a forward PINN reproduced the trajectories to 0.4–0.8%. However, (10) is a stable linear filter driven by j^2 : for *any* current waveform, integrating (10) forward produces a latent trajectory $z(\theta)$ whose residual is identically zero regardless of whether an arc is present. Consequently, $\hat{\tau}$ is unidentifiable from current alone (its estimated profile is flat), and a scale degeneracy couples the normalisations G and P_0 . Second, a coupled circuit–arc formulation was subjected to a pre-registered identifiability gate. The circuit side passed convincingly: the diode-bridge-plus-capacitor topology fitted all 22 no-arc recordings with normalised RMSE 0.04–0.18, against 0.61–0.83 for a resistive control; the mains source was eliminated analytically through $v_s'' + (2\pi)^2 v_s = 0$ and recovered to 3–6%; and the circuit residual rejected surrogate noise by factors of 91–143. The arc branch, in contrast, failed every criterion: collapsing the arc state produced 99.9% reconstruction error yet the optimiser escaped through $p_0 \rightarrow 0$, spurious arc conductance appeared on no-arc data, and substituting the Cassie form for the Mayr form changed the fit by only 17%. Therefore, arc-plasma equations are unidentifiable from current-only terminal measurements of this dataset, and any detector claiming otherwise would inherit an unverifiable physics term; Fig. 2 summarises the gate evidence.

F. Independent Verification of the Dynamic Model

Before any detector is constructed, (9) must earn its status as a dynamic model. A verification suite is executed under nested leave-one-SMPS-model-out (LOMO): (i) one-step prediction on held-out models against a persistence null ($\Delta z = 0$); (ii)

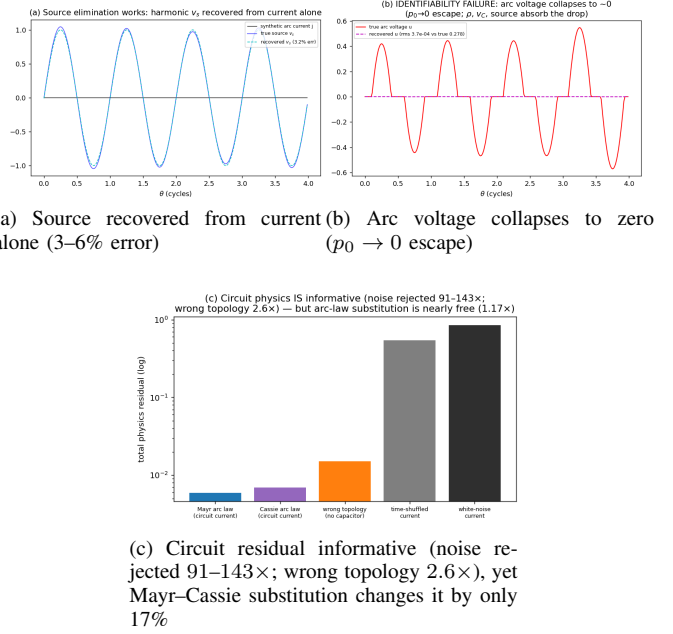


Fig. 2. Pre-registered identifiability gate for the coupled circuit–arc formulation: the circuit side passes, whereas the arc branch fails every criterion.

multi-step rollout at horizons of 1–16 half cycles; (iii) null-dynamics controls, namely a sign-flipped map $(-A, -c)$ and a map trained on temporally shuffled pairs; (iv) recording-level bootstrap identifiability (300 resamples) with library conditioning; and (v) behavioural consistency, requiring elevated residuals in all labelled arc recordings and no false anomalies on A4 and the supplementary units.

G. PINN Detector

The detector operates on a decision point per half cycle. Its inputs are the phase-binned raw current of the previous four estimated cycles (4×64 bins), the state sequence $z_{k-16:k}$, and the physics residual sequence

$$r_j = z_{j+1} - (I + A)z_j - c, \quad (11)$$

computed with the fold-specific frozen map. The raw input is the phase-binned current of the previous four estimated cycles: the trailing $\lfloor 4\hat{P} \rfloor$ samples are linearly interpolated onto a fixed grid of 4×64 phase bins and standardised per window,

$$\bar{v}_b = \frac{v_b - \mu_v}{\sigma_v + \epsilon}, \quad v = \text{interp}(x_{n-\lfloor 4\hat{P} \rfloor:n}, 256), \quad (12)$$

where $\text{interp}(\cdot, 256)$ denotes linear resampling of the trailing window onto 256 phase bins and μ_v, σ_v are the per-window mean and standard deviation; the waveform representation therefore carries no rate information in the rate-independent branch. Because the physics residuals (11) are heavy-tailed (arc residuals exceed no-arc residuals by up to two orders of magnitude, with $|r|$ reaching 89 standardised units), a signed-logarithmic compression conditions all physics channels,

$$\ell(x) = \text{sign}(x) \ln(1 + |x|/S_0), \quad S_0 = 0.5, \quad (13)$$

where S_0 is the pre-registered no-arc residual scale from the verification stage, not a tuned quantity. Without (13), training-set standardisation is dominated by arc outliers and numerically crushes the no-arc dynamics structure (its retained variance falls to 0.21), a pathology consistent with the established analyses of unbalanced PINN objectives, in which a subset of large residual terms dominates the gradient flow [31], [32]. The frozen map coefficients carry bounded trainable refinements,

$$\begin{aligned} A_{\text{eff}} &= A + 0.3 \tanh(\Delta A) \odot \max(|A|, 0.05), \\ c_{\text{eff}} &= c + 0.3 \tanh(\Delta c), \end{aligned} \quad (14)$$

where $\odot$ denotes the elementwise product and ΔA , Δc are trainable; the $\tanh$ parameterisation guarantees that the refinement never departs from the verified equation by more than 30%. The network comprises a one-dimensional convolutional encoder on (12), a separate multilayer branch on the ℓ -compressed state and residual sequences, a detection head producing the logit s_k , and a physics forecast head $\hat{z}_{k+1}$. Training minimises

$$\mathcal{L} = \text{BCE}_w(s_k, y_k) + \lambda \mathbb{I}[y_k=0] \left\| \hat{z}_{k+1} - (z_k + A_{\text{eff}} z_k + c_{\text{eff}}) \right\|^2, \quad (15)$$

where BCE_w is the class-weighted binary cross-entropy over the arc label y_k and the physics term supervises the representation toward the verified nominal dynamics on no-arc training windows only; the label gate $\mathbb{I}[y_k=0]$ acts in training exclusively, and inference is entirely label-free.

H. Evaluation Protocol, Ablations, and Benchmarks

A leakage-free nested LOMO protocol is enforced at the code level. For each outer held-out SMPS model, three inner folds are formed; the SINDy map, feature scalers, and all fitted objects for an inner fold use only the two inner-training models, and the inner-validation model contributes to nothing that is fitted. The PINN configuration is selected separately per outer fold from its inner folds only (equal eight-configuration budgets), after which the pipeline is refit on all three non-outer models and frozen. Throughout this paper, *frozen* denotes that every element of a pipeline—the architecture and selected hyperparameters, the three-model SINDy map, the training recipe and random seeds, the deployment threshold, and the Platt calibration parameters—is fixed using inner data only, before the corresponding outer SMPS model is examined for the first time, and nothing is altered afterwards; the outer test is therefore executed exactly once per pipeline with no possibility of test-informed adjustment. Deployment thresholds and probability calibration are fitted on pooled inner out-of-fold (OOF) predictions only: the threshold maximises the MCC over the OOF scores, and Platt scaling [41]

$$\tilde{p} = \sigma(ap + b), \quad (16)$$

maps the raw score p to a calibrated probability through the logistic function $\sigma(\cdot)$, with (a, b) fitted per outer fold by logistic regression on the OOF predictions. A programmatic leakage audit enumerating 76 fitted objects confirmed zero violations.

The matched comparisons, under identical inputs, architectures, parameter budgets, splits, seeds, and tuning budgets, are: the full model; a matched no-dynamics control (residual channels zeroed, physics loss off); residual-channels-only; physics-loss-only; and a wrong-dynamics-loss-only control in which the sign-flipped map replaces (9) in the loss with no residual channels supplied to either arm, so that the equation is the only difference. Contextual benchmarks are tuned on the same inner folds with comparable budgets: harmonic-order band features with logistic regression (FFT+LR) [37], gradient-boosted trees on harmonic and time-domain features (GBM) [35], a one-dimensional convolutional network (CNN), and a gated recurrent unit network (GRU) [36]. Average precision (PR-AUC) is the primary metric [39], complemented by AUROC, MCC [38], the Brier score [40], event detection rate, detection delay, and false alarms per 100 cycles; confidence intervals employ fold- and recording-level cluster bootstrap [42] with 10,000 resamples.

III. RESULTS

A. Setup

All experiments were conducted in Python 3.12 (PyTorch, scikit-learn, NumPy/SciPy) with deterministic per-fit seeds; approximately 6,300 model fits were executed and logged across all stages. The outer test was performed exactly once per frozen pipeline with 5 seeds for the PINN variants, FFT+LR, and GBM, and 3 seeds for the CNN and GRU; this deviation from the pre-specified 10 seeds is recorded and is benign given observed seed standard deviations below 0.012.

B. Verification of the Discovered Dynamics

Table II summarises the verification suite. The discovered map predicts held-out SMPS models better than persistence in 41/47 recordings and, decisively, at every rollout horizon from 1 to 16 half cycles, which excludes static or label-correlated artefacts. The sign-flipped map degrades prediction in 47/47 recordings, and shuffled-pair training degrades it in both branches, confirming that the correct temporal ordering is being learned. The diagonal contraction terms are identified with bootstrap support frequency 1.00 and confidence intervals excluding zero; the library condition number is approximately 22, and the results are flat across the STLSQ threshold grid. Arc-interval residuals exceed no-arc residuals in 21/21 labelled arc recordings with Cohen's $d = 2.0$ – 2.2 and disjoint confidence intervals, while A4 exhibits 0% anomalies and the supplementary normal units remain within the no-arc regime (5% of the arc residual level). The rate-independent and rate-aware coefficient estimates agree within their bootstrap intervals, as detailed in Table I.

It is worth mentioning that the causal trailing-median baseline in (5) is itself an audited correction: the initial fixed-calibration formulation was revised after the drift audit reported above (conducted before the supplementary units were examined), which improved held-out prediction and made the model beat persistence at every rollout horizon; the original result is documented rather than erased.

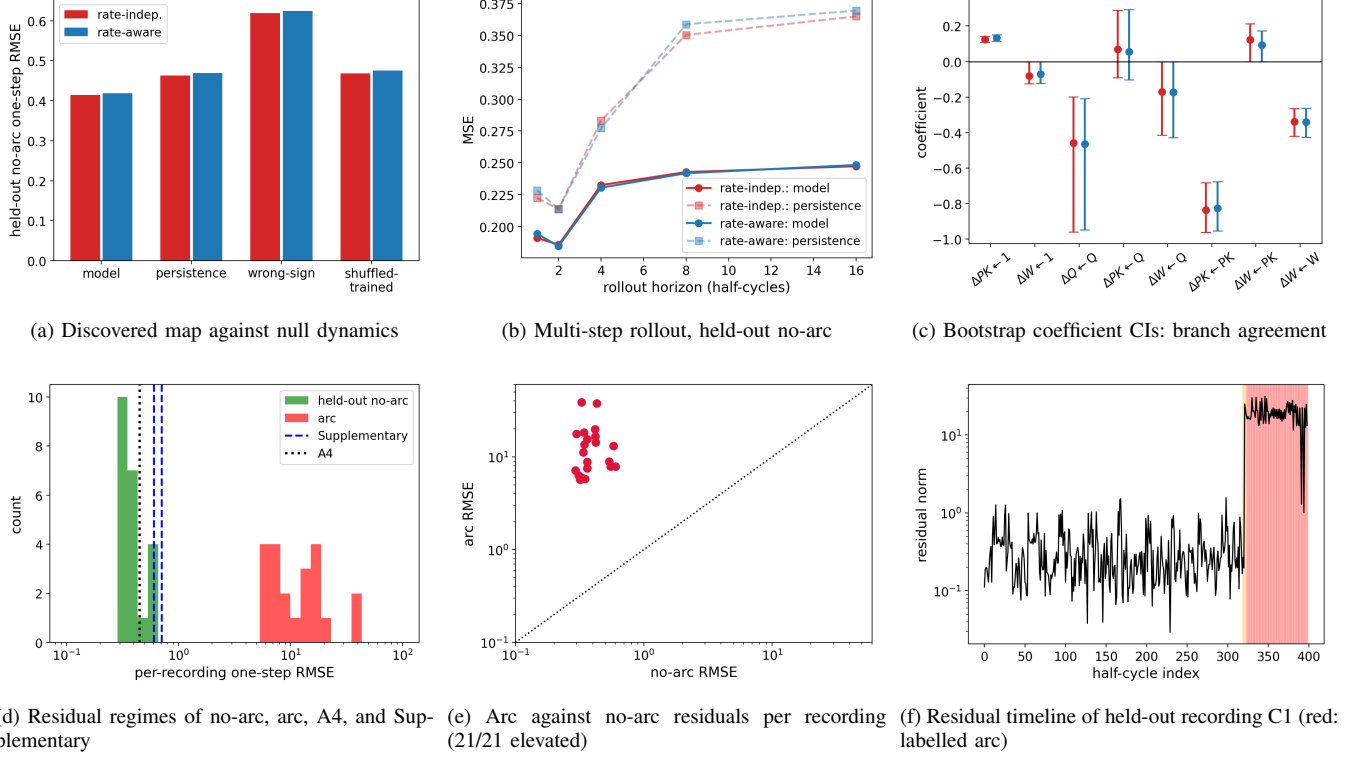


Fig. 3. Independent verification of the discovered device dynamics on held-out SMPS models under nested LOMO, for both sampling-rate branches.

TABLE I
IDENTIFIABILITY OF THE DISCOVERED MAP (RECORDING-LEVEL
BOOTSTRAP, 300 RESAMPLES; RATE-INDEPENDENT BRANCH).

Term	Mean	95% CI	Support freq.
$\Delta Q \leftarrow Q$	-0.46	[-0.95, -0.21]	1.00
$\Delta PK \leftarrow PK$	-0.83	[-0.95, -0.68]	1.00
$\Delta W \leftarrow W$	-0.34	[-0.43, -0.26]	1.00
Cross-terms	individually small		0.18–0.95
Library condition number	≈ 22		
STLSQ threshold sensitivity	no-arc RMSE 0.397–0.401 (flat)		

TABLE II
VERIFICATION OF THE DISCOVERED DEVICE DYNAMICS (HELD-OUT
SMPS MODELS; STANDARDISED UNITS).

Test	Rate-indep.	Rate-aware
One-step no-arc RMSE	0.415	0.419
vs. persistence null	0.463 (20/23)	0.469 (21/24)
vs. sign-flipped map	0.619 (23/23)	0.568 (24/24)
vs. shuffled-pair-trained	0.468	0.531
Rollout MSE, $h=16$ (model/persist.)	0.247/0.365	0.248/0.370
Arc-interval RMSE [95% CI]	13.2 [10.0, 16.9]	13.7 [10.2, 17.9]
No-arc RMSE [95% CI]	0.415 [0.37, 0.46]	0.419 [0.38, 0.47]
Effect size (Cohen's d)	2.18	2.02
Arc > no-arc per recording	21/21	21/21

C. Matched Ablations

Table III reports the pre-specified paired tests over 12 leakage-free inner folds and 5 seeds. With the baseline representation, removing the dynamics pathway degrades average precision by +0.073 [+0.033, +0.116] (rate-independent;

TABLE III
PRE-SPECIFIED PAIRED ABLATIONS, ΔAP [95% CLUSTER-BOOTSTRAP
CI].

Comparison	Rate-indep.	Rate-aware
Dynamics removed	+0.073 [+0.033, +0.116]	+0.049 [+0.014, +0.087]
Physics loss removed	+0.014 [+0.002, +0.030]	+0.015 [+0.003, +0.033]
Channels vs. none	+0.059 [+0.026, +0.096]	+0.034 [+0.007, +0.066]
Equation test (loss only)	-0.008 [-0.021, +0.000]	+0.001 (ns)
At ceiling (repaired encoder)	+0.0006 [-0.002, +0.003]	-0.0012 [-0.004, +0.001]

11/12 folds; standardised effect $d_z = 0.95$) and +0.049 [+0.014, +0.087] (rate-aware); the residual channels carry most of the benefit, and the physics loss contributes a smaller yet significant increment (+0.014/+0.015). In contrast, the valid equation-specificity test—correct against sign-flipped dynamics in the loss with no residual channels—is null in both branches. Consequently, the correctness of (9) rests on the predictive verification of Table II, where the flipped map fails decisively; no detector-level equation-sign specificity is claimed. An earlier sign-flip control with residual channels supplied is reported as inconclusive, the flipped residual being an affine-invertible recoding of information already present in the inputs.

D. Conditioning of the Physics Channels

A code-level audit revealed that the initial detector underperformed for implementation rather than physical reasons: the heavy-tailed residual channels ($|r|$ up to 89 in standardised units) caused training-set standardisation to compress

TABLE IV
INNER-VALIDATION COMPARISON (12 LEAKAGE-FREE FOLDS; PR-AUC).

Method	AP		Worst fold	MCC
	indep.	aware		
GBM (harmonic feat.)	0.997	0.997	0.968	0.84–0.91
PINN (repaired)	0.9971	0.9970	0.987	0.92–0.94
PINN, no dynamics	0.9965	0.9981	0.987	0.90–0.93
FFT+LR	0.989	0.990	0.923	0.73–0.74
CNN	0.872	0.878	0.44	0.85–0.87
PINN (before repair)	0.678	0.669	0.24	0.55–0.58
GRU	0.784	0.635	0.16	0.66–0.73

the no-arc dynamics structure to a variance of 0.21, and the fully connected encoder could not extract the waveform structure that a fixed spectral basis provides. Introducing the signed-logarithmic compression with the pre-registered scale and the convolutional encoder raised the validation average precision from 0.64 to 0.994–0.995 in both branches without altering the physics. This observation is consistent with the established finding that imbalanced residual terms can dominate and pathologically bias PINN optimisation [31]. At the resulting accuracy ceiling, the matched ablation is null (Table III, last row): a competent encoder extracts from the raw waveform everything that this dataset’s arcs offer, and the dynamics pathway becomes performance-neutral while retaining its interpretability. Table IV summarises the inner-validation performance of the repaired detector against the benchmarks under identical folds and comparable budgets. The repaired PINN attains an average precision of 0.9970–0.9971 with a worst fold of 0.987—the best worst-case of any method, including the GBM (0.968)—and, notably, resolves the hard SMPS model on which every learned baseline had previously failed (0.992–0.996, against 0.37 for the unrepaired detector, 0.49–0.51 for the CNN, and 0.21–0.22 for the GRU at validation on that model).

E. One-Shot Outer Test

Table VI presents the outer-test results with all pipelines frozen in the sense defined in Section II. The proposed PINN attains an average precision of 0.9924 (rate-independent) and 0.9954 (rate-aware), detects 105/105 labelled arc events with a median delay of zero half cycles, produces 0.59 false alarms per 100 cycles at the frozen thresholds, and achieves Brier scores of 0.024–0.026 with the frozen Platt calibration. The validation-to-outer optimism gap is approximately 0.004, indicating that the nested protocol was honest. It is worth emphasising the false-alarm column of Table VI: at the frozen thresholds, the proposed detector produces 0.59 false alarms per 100 cycles, against 2.0–5.6 for the GBM and 4.7–5.4 for FFT+LR, i.e., a 3–10 $\times$ advantage over the accuracy leader on precisely the axis—nuisance tripping—that limits the field acceptance of commercial arc-fault detection devices. One can notice that the outer test reversed the inner-validation ranking below the top: the FFT+LR detector, dominant at validation, collapsed on the held-out hard model (average precision 0.48–0.57 on Model A), as did the CNN and GRU, whereas the frozen PINN retained 0.991 on the same

TABLE V
OUTER-TEST PR-AUC PER HELD-OUT SMPS MODEL.

Branch	Method	A	B	C	D
indep.	PINN (proposed)	0.9913	0.9788	1.000	0.9994
indep.	PINN, no dyn.	0.9741	0.9972	1.000	0.9993
indep.	GBM	0.9995	0.9991	1.000	1.000
indep.	FFT+LR	0.5718	1.000	1.000	1.000
aware	PINN (proposed)	0.9913	0.9909	1.000	0.9996
aware	PINN, no dyn.	0.9929	0.9992	1.000	0.9992
aware	GBM	0.9995	0.9970	1.000	1.000
aware	FFT+LR	0.4776	1.000	1.000	1.000

TABLE VI
ONE-SHOT OUTER TEST WITH FROZEN PIPELINES, I.E., ALL CONFIGURATIONS, MAPS, THRESHOLDS, AND CALIBRATIONS FIXED FROM INNER DATA BEFORE THE OUTER MODELS WERE EXAMINED (MEAN OVER FOUR HELD-OUT SMPS MODELS).

Method	AP		Worst model	Events	Delay (med.)	FA/100 cycles
	indep.	aware				
GBM	0.9997	0.9991	0.997	105/105	0	2.0–5.6
PINN (proposed)	0.9924	0.9954	0.979/0.991	105/105	0	0.59
PINN, no dynamics	0.9927	0.9978	0.974/0.993	105/105	0	0.35
CNN	0.903	0.898	0.59–0.61	63/63	0–0.5	0.0
FFT+LR	0.893	0.869	0.48–0.57	105/105	0	4.7–5.4
GRU	0.824	0.802	0.30–0.32	49–53/63	0–2	0.3–5.0

model in both branches. The GBM benchmark matched or exceeded the PINN (0.9991–0.9997); this is reported plainly, as the proposed method is not required to outperform every benchmark, and the GBM inherits its rate independence from the proposed phase-binned representation. The paired outer difference between the full PINN and its matched ablation is $-0.0003/-0.0024$, confirming the ceiling-level null on outer data. Table V decomposes the outer results per held-out model, which makes the generalisation asymmetry explicit: on Models B–D every strong method is near-saturated in both branches, whereas Model A separates the robust methods (PINN 0.9913, GBM 0.9995) from the collapsed ones (FFT+LR 0.478–0.572, CNN 0.59–0.61).

1) *Hierarchical Statistics and Non-Inferiority*: Because recordings are clustered within SMPS models, a recording-level hierarchical bootstrap (2,000 resamples over the pooled outer predictions) complements the per-model means: the pooled outer average precision of the proposed detector is 0.961 [0.925, 0.994] in the rate-independent branch and 0.992 [0.986, 0.997] in the rate-aware branch. The pre-specified non-inferiority test (margin 0.01 on the paired per-model differences) yields: proposed PINN, $\Delta = -0.0031$ with confidence bound -0.0090 , **non-inferior**; GBM, $+0.0005$ (-0.0005), non-inferior; the no-dynamics PINN (-0.0052 ; -0.0141) and FFT+LR ($+0.0235$; -0.0226) are inconclusive at this margin. Therefore, withholding the sampling rate is statistically established as cost-free for the proposed detector, not merely observed.

F. Robustness to Sampling-Rate Changes

To assess sampling-rate robustness, the outer recordings were anti-alias resampled to 4.8 kHz and 3.2 kHz with exactly remapped labels, and the frozen pipelines were applied without

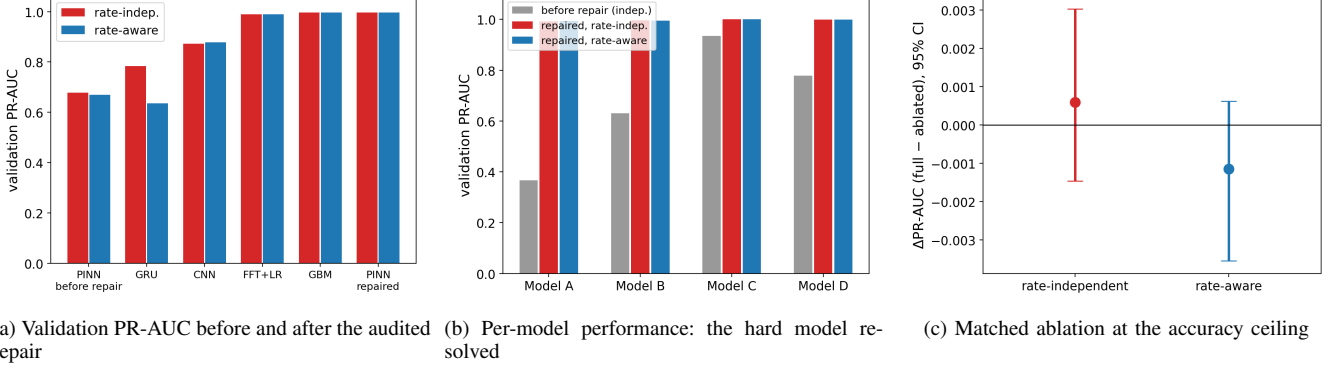


Fig. 4. Repair of the detector through the audited conditioning fixes (physics unchanged) and the resulting ceiling-level matched ablation with 95% cluster-bootstrap confidence intervals.

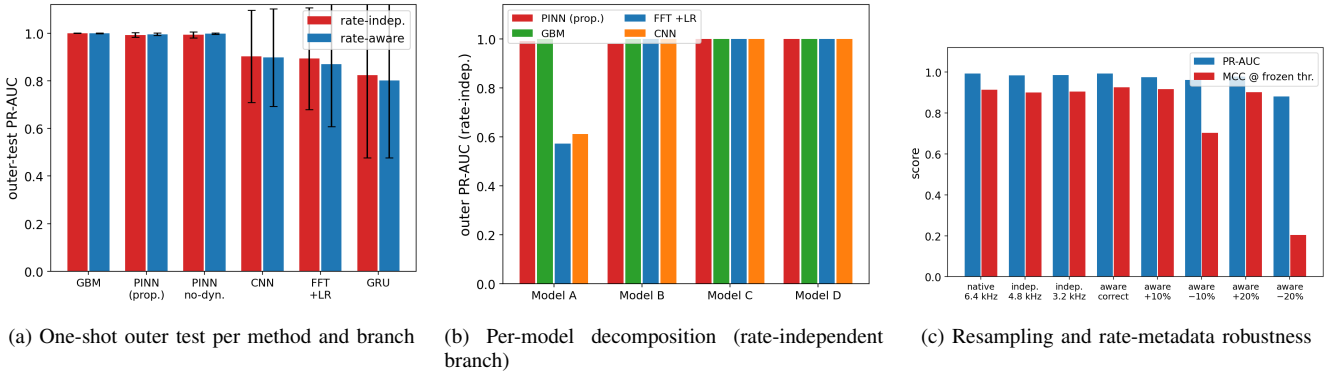


Fig. 5. One-shot outer-test evaluation with frozen pipelines and the sampling-rate robustness study: the proposed detector is robust on the hard held-out model where the spectral, convolutional, and recurrent baselines collapse, and it retains its accuracy under resampling with no rate information, whereas rate-metadata errors severely degrade the rate-aware alternative.

modification. As illustrated in Table VII, the rate-independent detector, given no rate information whatsoever, retains an average precision of 0.9844 and 0.9853 at the two rates (native: 0.9924) with MCC of 0.90. The rate-aware detector performs comparably when supplied the correct new rate (0.9933) but degrades sharply under metadata errors: a -20% rate error reduces the mean MCC to 0.204, and on Model A to -0.008 . The degradation under metadata error is furthermore model-dependent: at -20% , the per-model MCC of the rate-aware detector falls to -0.008 (Model A), 0.559 (B), 0.097 (C), and 0.167 (D), i.e., three of the four models become effectively unprotected at the frozen threshold, while the corresponding average precision (0.685–0.984) would misleadingly suggest a functioning detector; threshold-level and ranking-level robustness must therefore be reported jointly. These results constitute the practical case for rate independence: the proposed pipeline removes an entire failure mode of deployed detectors at essentially no accuracy cost.

G. Normal-Control False Alarms

At the frozen thresholds, the no-arc-only recording A4 produces 0% false alarms (rate-aware branch; the causal branch has no A4 coverage because its warm-up consumes the short record, which is recorded as a limitation). On the two never-

TABLE VII
RESAMPLING ROBUSTNESS OF THE FROZEN PIPELINES (OUTER MODELS).

Case	AP (mean)	MCC @ frozen thr.
Native 6.4 kHz	0.9924	0.914
Rate-indep., 4.8 kHz, no rate given	0.9844	0.900
Rate-indep., 3.2 kHz, no rate given	0.9853	0.904
Rate-aware, 4.8 kHz, correct rate	0.9933	0.925
Rate-aware, rate error -10%	0.9609	0.703
Rate-aware, rate error -20%	0.8803	0.204

trained supplementary units, the false-alarm rates are 3.5–4.2% and 11.1–11.2% on average across the four frozen pipelines (range 0–31%), although the underlying residuals remain an order of magnitude below arc levels. Therefore, per-device threshold calibration is identified as a deployment requirement for unseen hardware, quantified here rather than concealed.

H. Computational Performance

Offline, each PINN training run (18 epochs on approximately 5,200 decision points) required 8–10 s on a 4-thread CPU container. Online, inference is a single forward pass of a 25k–38k-parameter network per half cycle over causally computed features, with the period estimator updating every 64 samples from a 1536-sample trailing buffer; the pipeline is

therefore compatible with embedded-class real-time execution, although hardware-in-the-loop validation is left to future work.

IV. DISCUSSION

A. When Does the Physics Help?

The matched ablations of Table III localise the value of the discovered dynamics precisely. With a representation-limited detector, the physics residual channels contribute a statistically supported improvement of $+0.049$ to $+0.073$ average precision, and the physics forecast loss adds a further significant increment; at the accuracy ceiling of the repaired detector, the same pathway is performance-neutral within tight intervals (± 0.003). This is not a contradiction but a regime characterisation: the residual (11) is an engineered summary of temporal structure that a weak encoder cannot extract on its own, and a sufficiently expressive encoder trained on the same waveforms recovers equivalent information. It can be noticed that the equation-specificity test is null at the detector level while the same sign-flipped equation fails decisively at the predictive level (Table II); the correctness of the physics therefore lives in the system-identification evidence. In deployment terms, the validated model supplies a meaningful explanation for each alarm—a quantified violation of the device’s verified charge-balance dynamics—which a boosted-tree score cannot.

B. Why the Outer Test Reversed the Validation Ranking

The collapse of FFT+LR, the CNN, and the GRU on the held-out Model A, despite dominant or competitive validation scores, follows from the nesting itself: the per-outer configurations are selected without ever observing the outer model, so methods whose hyperparameters index device-specific spectral structure transfer poorly, whereas the proposed detector’s phase-binned representation with physics-conditioned channels transferred robustly (0.9913 on the same model). The GBM’s robustness is attributable to largely device-invariant harmonic-ratio features; it is worth mentioning that these are computed from the proposed rate-free phase-binned windows, so the strongest benchmark inherits its rate independence from the proposed pipeline. This reversal underlines the necessity of one-shot outer testing.

C. Threats to Validity

Four threats are recorded. First, the dataset comprises four SMPS models from a single acquisition campaign at one native rate; resampling emulates rate diversity but not sensor diversity. Second, the outer evaluation used 5 seeds (3 for the CNN and GRU) instead of the pre-specified 10; the observed seed variability (≤ 0.012) renders this benign. Third, thresholds transferred to entirely unseen devices over-fire (up to 11% on a supplementary unit), so the reported false-alarm rates hold for calibrated devices; a short per-device commissioning period suffices in principle. Fourth, the topology support for the interpretation of (9) is waveform-level, since device schematics are unavailable; the interpretation is stated as consistency, not derivation.

V. CONCLUSIONS

Nonlinear SMPS loads and heterogeneous acquisition platforms pose several challenges to reliable series arc-fault detection. In this work, we proposed a framework that discovers, verifies, and embeds the nominal device dynamics of the SMPS input stage into a current-only, voltage-free, sampling-rate-independent PINN detector. The discovered contraction map was validated as a genuine dynamic model through prediction, rollout, null controls, identifiability, and transfer, while the conventional arc-plasma equations were shown to be unidentifiable from current-only measurements and were rejected with documented evidence. The one-shot outer test demonstrated an average precision of 0.9924 without any sampling-rate knowledge, detection of all 105 labelled arc events with zero median delay, and robustness to resampling where rate-metadata errors break the rate-aware alternative. The matched ablations characterised the role of the physics honestly: the discovered dynamics improve detection when the learned representation is weak and are performance-neutral at the accuracy ceiling, and a gradient-boosting benchmark on harmonic features matches the proposed detector on this dataset. A limitation of this work is that thresholds transferred to entirely unseen device models produce elevated false alarms (up to 11% on the supplementary units), which necessitates per-device threshold calibration; furthermore, the dataset comprises four SMPS models from a single acquisition campaign, and the causal branch lacks coverage of very short records. In future research, the discovered map will be developed as a grey-box model with component-level priors, the per-device calibration will be automated from a short commissioning period, and the framework will be validated on hardware-in-the-loop arc generation platforms across additional load classes; furthermore, since the nominal dynamics are discovered and calibrated from healthy data alone, the physics-residual pathway will be developed into a novelty detector for fault types absent from any training signature.

ACKNOWLEDGMENT

The authors extend their appreciation to Prince Sattam bin Abdulaziz University for funding this research work through the project number (PSAU/2025/01/34748). The authors also thank Chint and the Future Energy Laboratory for their support of this work.

APPENDIX

A. Pre-Registered Constants and Tuning Budgets

All constants were fixed before the experiments that use them: STLSQ threshold 0.05 and linear library (label-free sensitivity, flat over the grid); network width 128; compression scale $S_0 = 0.5$ (the verification-stage no-arc residual level); calibration length 30 half cycles with floored robust scales; trailing-baseline window 60 half cycles (drift audit); estimator lag range 16–512, buffer 1536, stride 64 samples; context length 16 half cycles. Tuning used equal eight-configuration budgets per method arm over learning rate $\{10^{-3}, 3 \times 10^{-3}\}$, encoder channels $\{16, 32\}$, and physics weight $\lambda \in \{0.3, 1.0\}$ or dropout $\{0.1, 0.3\}$ for arms without a physics loss; final

training used 18 epochs (20 for the CNN and GRU benchmarks). Approximately 6,300 fits were executed and logged in total across the formulation comparisons, ablation rounds, benchmarks, repaired-detector round, threshold calibration, and the outer test with robustness cases; every fit's seed, configuration, fold, and metrics are recorded in machine-readable logs, and a SHA-256 manifest covers the complete study package.

B. Data Division

For each outer fold, the three non-outer SMPS models form the development set, with three inner folds each training on two models and validating on the third; equation discovery, configuration selection, thresholds, and calibration use inner data only, the supplementary units never enter any fitted object, and the outer model is evaluated exactly once with everything frozen. A programmatic audit of all 76 fitted objects confirmed zero violations.

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